



Model Name & Configs

Llama-3.1-70B,
FlashInfer, vLLM, TP=4

Input Workload

(e.g., ShareGPT Dataset)

1 Emulated Serving Engine

Prompts

Native Memory
Abstraction

Native
Scheduler

Batch

State
Update

2 Step-time Predictor

Model-to-OpSet

$\Sigma (\text{count} \times \text{latency})$



TTFT / TPOT
Throughput
Queue & KV Usage

Offline:

DOOLY

Latency
DB

RandomForest
Estimator

Prediction
Tables